Technical Communication & Documentation Workflows

A Practical Guide to Modern Engineering Tools

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Presentation Roadmap

1 Distributed Version Control

2 Scientific Typesetting

3 Automated API Documentation

4 Cloud Documentation

Git & GitHub: Tracking and Collaboration

Local Git Lifecycle

- `git init`: Initialize repository
- `git add`: Stage workspace edits
- `git commit`: Record snapshots

Remote Collaboration

- Branching for isolated features
- Merging & conflict resolution
- Pull requests & release tags

Structured Authoring with LaTeX & Beamer

Why Plain-Text Typesetting?

Separating document logic from visual layout ensures consistent formatting across research papers and presentations.

- **Mathematical Precision:** Native rendering for complex equations such as $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.
- **Automated Indexing:** Dynamic Table of Contents, cross-references, and BibTeX citations.
- **Beamer Decks:** Modular frames, themes, and structured visual blocks.

Source-to-Manual Generation with Doxygen

- Parses annotated C/C++ header and source files into structured HTML and PDF references.
- Eliminates drift between codebase updates and external manuals.

Standard Javadoc-Style Tags

`@file` (module overview) • `@brief` (summary)
`@param` (input arguments) • `@return` (output values)

Collaborative Editing in Google Docs

- **Semantic Hierarchy:** Utilizing Title, H1, and H2 styles for navigable outlines.
- **Asynchronous Review:** Suggesting mode, anchored comment threads, and version history.
- **Archival Exports:** Generating clean, hyperlink-preserving PDF deliverables.